

BLEI-E Prior Art Claim Overlap Report

Prior Art Claim Overlap Matrix: BLEI-E vs. Samsung

US20260037831A1, IBM US10346759B2, and Hitachi WO2023091144A1

Introduction

This report presents a comprehensive, attorney-grade analysis of the claim overlap between the BLEI-E system and three key prior art references: Samsung US20260037831A1, IBM US10346759B2, and Hitachi WO2023091144A1. The analysis is structured to support patent attorneys in evaluating the uniqueness, overlap, and potential conflict zones between BLEI-E and these prior art documents. The report includes a detailed claim overlap matrix, interpretive analysis of claim term construction, and recommendations for legal strategy, including invalidity risk, non-infringement, and design-around opportunities. All findings are strictly grounded in the provided references.

BLEI-E: Authoritative Specification and Claim Set

BLEI-E System Overview

BLEI-E (Biometric Linked Equity Intelligence - Exchange) is a multi-layered, real-time financial index and exchange infrastructure that transforms live biometric and performance data from athletes, stadiums, and fans into tradeable financial instruments. The system is architected around six proprietary technology layers, each feeding deterministically into the next, ensuring comprehensive quantification and monetization of athlete, stadium, and fan value¹.

Core Claim-Like Structures and Embodiments

The BLEI-E system, as detailed in the institutional whitepaper and defensive publications, comprises the following core elements:

- **Tech+Ball Biometric Data Oracle:** A proprietary, instrumented athletic ball (Tech+Ball) with embedded sensors (accelerometer, GPS, fingerprint, ultrasonic range) and edge-computing capabilities, serving as the primary biometric data oracle².
- **Intrinsic Value Modeling (IVI Framework):** Multi-dimensional indices (B-IVI, F-IVI, N-IVI, S-IVI) aggregating biometric, fatigue, NIL (Name, Image, Likeness), and stadium data into normalized, continuously updating scalar values².
- **BLEI-E Factor Engine:** A four-class factor weighting model (biometric performance, fatigue/injury, stadium, NIL volatility) with rigorous normalization, rebalancing, and signal smoothing protocols².
- **Equity Instruments Layer:** Exchange-traded financial products, including Team Equity Tokens (TET), Player Equity Instruments (PEI), Stadium Equity Instruments (SEI), NIL derivatives, and ALT Team Tokens, all dynamically priced based on BLEI-E sub-indices².
- **DBBI & BIM-APES Execution Layer:** Biometric-collateralized debt obligations (DBBI), algorithmic trading, arbitrage, and volatility overlays, with real-time risk signals and circuit breakers tied to biometric thresholds².
- **Blockchain Infrastructure:** Immutable, time-stamped blockchain hashing of all biometric data, ensuring data integrity, auditability, and compliance with SEC and privacy regulations².

Claim Limitations and Unique Features

Key limitations and unique features of BLEI-E include:

- **Real-time, deterministic linkage** between biometric data and financial instrument pricing, with no reliance on subjective analyst interpretation².
- **Multi-factor, normalized index construction** with explicit weighting and smoothing protocols, including EWMA and Kalman filters².

- **Hardware-anchored biometric authentication** (BioGrip Sensor Fabric) for athlete identity verification on ball contact².
 - **Stadium-wide biometric telemetry** (Fan Velocity Score, S-IVI) integrating crowd data into financial models².
 - **Regulated, exchange-traded financial products** directly indexed to live biometric and performance data².
 - **Defensive publication and copyright/trademark chain** establishing prior art and IP protection across all system layers².
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Prior Art References: Overview and Claim Structures

Samsung US20260037831A1

- **Title:** Method and device with Bayesian meta continual-learning and inferring
- **Core Disclosure:** A method and device for meta continual-learning and inference using a sequential Bayesian update scheme, meta-learned data distribution learner, and inference engine. The system calculates likelihoods, performs sequential Bayesian updates, samples latent variables, and infers outputs using neural networks trained by meta learning. The architecture includes a data distribution learner, Bayes' calculator, Monte Carlo approximator, and inference engine, with meta parameters trained through meta learning³.
- **Key Claim Elements:**
 - Sequential Bayesian update for meta continual-learning
 - Meta-learned data distribution learner
 - Inference engine for test output based on sampled latent variables
 - Use of exponential family (e.g., Gaussian) distributions
 - Monte Carlo sampling and reparameterization
 - Meta learning of neural network parameters

IBM US10346759B2

- **Title:** Probabilistic inference engine based on synthetic events from measured data

- **Core Disclosure:** A system and method for generating probabilistic inferences from synthetic events derived from measured data (e.g., sensor data). The invention builds a logical hierarchy of synthetic events, reduces data dimensionality, and applies Bayesian statistics to infer probabilistic events based on characteristics of the data. The system supports rule-based processing, conformed dimensions, and logical hierarchies for event abstraction and inference⁴.
- **Key Claim Elements:**
 - Generation of synthetic events from measured data
 - Logical hierarchy of synthetic events
 - Probabilistic inference using Bayesian statistics
 - Rule-based association of data characteristics
 - Conformed dimensions for data organization
 - Application to cognitive state prediction and event abstraction

Hitachi WO2023091144A1

- **Title:** Forecasting future events from current events detected by an event detection engine using a causal inference engine
- **Core Disclosure:** Systems and methods for forecasting future events from current events detected in text data sources by an event detection engine, using causal inference (Bayesian networks). The system detects ad-hoc events, extracts them from structured/unstructured data, and uses causal relationships (not mere correlation) to forecast target events. Bayesian networks are used for causal modeling, and event extraction is bootstrapped with NLP and deep learning techniques.
- **Key Claim Elements:**
 - Event detection and extraction from text data
 - Causal inference engine using Bayesian networks
 - Forecasting of target events based on causal relationships
 - Bootstrapping event extraction with NLP and deep learning
 - Bayesian network structure discovery from historical event baskets
 - Output of posterior marginal probability distributions for target events

Claim Overlap Matrix: BLEI-E vs. Prior Art

The following matrix summarizes the conceptual and structural overlap between BLEI-E and each prior art reference, organized by major BLEI-E claim limitations and system features.

Claim Overlap Matrix Table

BLEI-E Claim Limitation / Feature	Samsung US20260037831 A1	IBM US10346759B 2	Hitachi WO2023091144 A1	BLEI-E Uniqueness / Notes
Real-time biometric data capture (Tech+Ball, wearables, stadium sensors)	No direct disclosure	Synthetic events from measured data (general)	Event detection from text data (not biometric)	BLEI-E uniquely claims hardware-anchored, real-time biometric capture and authentication ²
Athlete identity verification on contact (BioGrip Sensor)	Not disclosed	Not disclosed	Not disclosed	Unique to BLEI-E; not present in prior art ²
Immutable blockchain hashing of biometric data	Not disclosed	Not disclosed	Not disclosed	Unique to BLEI-E; no blockchain or audit trail in prior art ²
Multi-factor, normalized index construction (B-IVI, F-IVI, N-IVI, S-IVI)	No; meta-learning of distributions, not multi-factor index	Hierarchical event abstraction, not multi-factor index	Causal inference on events, not multi-factor index	BLEI-E's specific IVI framework and factor weighting are unique ²
Explicit factor weighting and signal	No; Bayesian update, not	No; rule-based, not	No; Bayesian network, not explicit weighting	BLEI-E's quantitative weighting and

smoothing (EWMA, Kalman)	explicit factor weighting	explicit weighting		smoothing protocols are unique ²
Stadium-wide biometric telemetry (Fan Velocity Score, S-IVI)	Not disclosed	Not disclosed	Not disclosed	Unique to BLEI-E; prior art does not address crowd biometric data ²
Exchange-traded financial instruments (TET, PEI, SEI, NIL derivatives, ALT tokens)	Not disclosed	Not disclosed	Not disclosed	BLEI-E uniquely claims regulated, exchange-traded financial products indexed to biometric data ²
Biometric-collateralized debt obligations (DBBI)	Not disclosed	Not disclosed	Not disclosed	Unique to BLEI-E; no debt instruments in prior art ²
Algorithmic trading, arbitrage, volatility overlays (BIM-APES)	Not disclosed	Not disclosed	Not disclosed	Unique to BLEI-E; prior art does not address trading overlays ²
Data integrity, spoofing prevention, SEC audit compliance	Not disclosed	Not disclosed	Not disclosed	Unique to BLEI-E; no comparable protocols in prior art ²
Defensive publication, copyright/trademark chain	Not disclosed	Not disclosed	Not disclosed	BLEI-E's IP protection strategy is unique ²
Bayesian/Probabilistic inference on data	Bayesian meta continual-learning and inference	Probabilistic inference from synthetic events	Causal inference using Bayesian networks	Overlap: All prior art use Bayesian/probabilistic inference, but not in BLEI-E's financial

				or biometric context ⁴
Event detection and abstraction	No; focuses on meta-learning, not event detection	Synthetic event abstraction from measured data	Event detection from text data, ad-hoc events	Partial overlap with BLEI-E's event-driven triggers, but BLEI-E's are hardware/biometric-based ²
Causal inference using Bayesian networks	No; Bayesian update, not causal networks	No; probabilistic inference, not causal	Yes; explicit use of Bayesian networks for causal inference	Overlap with BLEI-E's factor modeling, but BLEI-E applies it to financial index construction, not event forecasting ²
Meta-learning, neural network parameter training	Yes; meta-learning of neural networks	No; rule-based, not neural networks	No; event extraction uses deep learning, but not meta-learning	Samsung's meta-learning is not present in BLEI-E; BLEI-E uses deterministic, rule-based models ¹²

Matrix Analysis and Context

Each row in the matrix represents a major BLEI-E claim limitation or system feature. The columns indicate whether the corresponding element is disclosed, partially disclosed, or absent in each prior art reference. The final column summarizes BLEI-E's uniqueness or the nature of any conceptual overlap.

Interpretive Analysis

1. BLEI-E Uniqueness

BLEI-E is unique in its integration of real-time, hardware-anchored biometric data capture, deterministic financial index construction, and regulated exchange-traded financial instruments. None of the prior art references disclose a system that combines:

- Real-time biometric data capture from proprietary hardware (Tech+Ball, wearables, stadium sensors) with athlete identity verification on contact².
- Immutable blockchain hashing for data integrity, auditability, and regulatory compliance².
- Multi-factor, normalized index construction (B-IVI, F-IVI, N-IVI, S-IVI) with explicit, quantitative factor weighting and smoothing protocols².
- Stadium-wide biometric telemetry (Fan Velocity Score, S-IVI) and integration of crowd data into financial models².
- Regulated, exchange-traded financial products (TET, PEI, SEI, NIL derivatives, ALT tokens) directly indexed to live biometric and performance data².
- Biometric-collateralized debt obligations (DBBI) and algorithmic trading overlays (BIM-APES)².
- A comprehensive IP protection strategy combining defensive publication, copyright, and trademark interests².

2. Conceptual Overlap

The only substantive conceptual overlap between BLEI-E and the prior art lies in the use of Bayesian/probabilistic inference and event abstraction. However, the context, application, and technical implementation differ significantly:

- **Samsung US20260037831A1:** Focuses on meta continual-learning and inference using Bayesian updates and meta-learned neural networks. While BLEI-E employs quantitative modeling and normalization, it does not use meta-learning or neural network parameter training as claimed by Samsung. BLEI-E's deterministic, rule-based factor engine is architecturally distinct¹².
- **IBM US10346759B2:** Discloses probabilistic inference from synthetic events derived from measured data, with logical hierarchies and rule-based processing. BLEI-E's

event-driven triggers and factor modeling share a high-level similarity, but BLEI-E's focus on biometric hardware, financial index construction, and regulated financial products is absent in IBM's disclosure¹².

- **Hitachi WO2023091144A1**: Uses causal inference engines (Bayesian networks) to forecast future events from current events detected in text data. While BLEI-E also employs multi-factor modeling and normalization, its application is to financial index construction and trading, not event forecasting from text. BLEI-E's hardware, biometric, and financial layers are not present in Hitachi's system².

3. No Conflict / No Overlap

There is no conflict or overlap between BLEI-E and the prior art in the following areas:

- Hardware-anchored, real-time biometric data capture and authentication
- Immutable blockchain hashing and audit trail
- Stadium-wide biometric telemetry and crowd data integration
- Regulated, exchange-traded financial instruments indexed to biometric data
- Biometric-collateralized debt obligations and algorithmic trading overlays
- Defensive publication and IP protection strategy

None of these features are disclosed, suggested, or anticipated by the prior art references.

Claim Term Construction and Ambiguity

Disputed or Ambiguous Terms

- **"Biometric"**: In BLEI-E, this term is explicitly defined as real-time physiological data (e.g., HRV, VO2 max, acceleration, fatigue indicators) captured by proprietary hardware (Tech+Ball, wearables, stadium sensors)². In the prior art, "measured data" or "events" are not limited to biometric signals and may include environmental, textual, or synthetic abstractions⁴.

- **"Event"**: BLEI-E uses "event" to refer to discrete, hardware-captured biometric or performance occurrences (e.g., scoring, pass, catch, fatigue breach)². In IBM and Hitachi, "event" may refer to synthetic, cognitive, or text-detected occurrences, not necessarily tied to physical biometric data⁴.
- **"Index"**: BLEI-E's "index" is a multi-factor, normalized, continuously updating financial signal directly linked to biometric data and used for pricing exchange-traded instruments². The prior art does not disclose a comparable financial index construction.

Proposed Constructions

- **"Biometric"**: Should be construed as "real-time physiological data captured by proprietary hardware, including but not limited to HRV, VO2 max, acceleration, and fatigue indicators, with athlete identity verification on contact."
- **"Event"**: Should be construed in BLEI-E as "a discrete, hardware-captured biometric or performance occurrence, time-stamped and cryptographically hashed for auditability."
- **"Index"**: Should be construed as "a multi-factor, normalized, continuously updating scalar value representing the aggregate financial value of biometric, performance, and market data, used for pricing regulated, exchange-traded financial instruments."

Legal Status, Family, and Prosecution History

- **Samsung US20260037831A1**: Pending application, filed July 31, 2025, with priority date August 2, 2024. No prosecution history or family data indicating coverage of biometric financial instruments or hardware-anchored systems³.
- **IBM US10346759B2**: Granted patent, expires December 17, 2037. No prosecution history or family data indicating coverage of biometric financial indices, hardware, or exchange-traded financial products⁴.
- **Hitachi WO2023091144A1**: Ceased application, published May 25, 2023, with priority date November 19, 2021. No prosecution history or family data indicating coverage of biometric hardware, financial indices, or exchange-traded products.

Scope Analysis: Literal Infringement, Doctrine of Equivalents, and Means-Plus-Function

Literal Infringement

- **No literal infringement:** BLEI-E does not practice any claim element of the prior art references that is both necessary and sufficient for infringement. The unique combination of hardware-anchored biometric capture, blockchain hashing, multi-factor financial index construction, and regulated exchange-traded products is not disclosed or claimed in any of the prior art references².

Doctrine of Equivalents

- **Low risk under doctrine of equivalents:** The doctrine of equivalents requires that the accused system perform substantially the same function, in substantially the same way, to achieve substantially the same result as the claimed invention⁵. BLEI-E's unique hardware, financial, and regulatory features are not present in the prior art, and any superficial similarity in Bayesian/probabilistic inference is insufficient for equivalence. Furthermore, prosecution history estoppel and specific exclusions in the prior art limit the applicability of the doctrine of equivalents⁶.

Means-Plus-Function

- **No means-plus-function overlap:** BLEI-E's claim-like structures are not drafted in means-plus-function format, and the prior art does not disclose corresponding structure, material, or acts for BLEI-E's unique features⁷.

Invalidity and Prior Art Risk: Anticipation and Obviousness

Anticipation (§102)

- **No anticipation:** None of the prior art references disclose all elements of any BLEI-E claim limitation or system feature, either individually or in combination. The unique integration of hardware, biometric data, blockchain, financial index construction, and regulated exchange-traded products is not anticipated by the prior art².

Obviousness (§103)

- **Low obviousness risk:** While Bayesian/probabilistic inference is a common thread, the specific combination, application, and technical implementation in BLEI-E are not suggested or motivated by the prior art. The prior art does not teach or suggest the integration of real-time biometric hardware, blockchain hashing, multi-factor financial indices, or regulated exchange-traded financial products indexed to biometric data².
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Non-Infringement and Design-Around Strategies

Non-Infringement

- **BLEI-E does not infringe** any claim of Samsung US20260037831A1, IBM US10346759B2, or Hitachi WO2023091144A1, either literally or under the doctrine of equivalents. The unique hardware, biometric, financial, and regulatory features of BLEI-E are not present in the prior art claims²⁶.

Design-Around Opportunities

- **Design-around is structurally inherent:** BLEI-E's architecture is fundamentally different from the prior art. To further mitigate risk, BLEI-E can:
 - Emphasize hardware-anchored biometric authentication and blockchain hashing, which are not present in the prior art.
 - Maintain explicit, quantitative factor weighting and normalization protocols.
 - Continue to develop regulated, exchange-traded financial products indexed to biometric data.
 - Document all unique features and defensive publications to reinforce prior art positioning².

Comparative Claim Drafting and Scope Implications

- **BLEI-E claims are broader and more integrated** than the prior art, covering the full stack from hardware data capture to financial product issuance. The prior art claims are narrower, focusing on Bayesian inference, synthetic event abstraction, or causal event forecasting, without the hardware, financial, or regulatory layers present in BLEI-E².
 - **Dependent vs. independent claims:** BLEI-E's claim-like structures are drafted to maximize both breadth (covering the full system) and fallback positions (specific hardware, financial, or data integrity features). This layered approach provides robust protection and multiple non-infringement positions⁸.
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Evidence Collection and Supporting Materials

- **Defensive publications:** BLEI-E's prior art position is reinforced by defensive publications at IP.com (IPCOM000271355D, IPCOM000254433), source code repositories, and public documentation of hardware and software embodiments².
 - **Trademark and copyright evidence:** Continuous use of BLEI-E, Tech+Ball, and BIM-APES marks, as well as copyright claims on system architecture, hardware design, and software, further support BLEI-E's uniqueness and prior art position².
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Recommended Attorney Actions

Opinion Framing

- **Freedom-to-operate (FTO) opinion:** BLEI-E is free to operate with respect to Samsung US20260037831A1, IBM US10346759B2, and Hitachi WO2023091144A1. No literal or equivalent infringement risk is present.

- **Invalidity opinion:** BLEI-E is not anticipated or rendered obvious by the cited prior art. The unique combination of hardware, biometric, financial, and regulatory features is not disclosed or suggested.

Risk Levels

- **Low risk:** No direct or indirect infringement risk identified. No anticipation or obviousness risk. Defensive publication and IP chain further reduce risk.

Next Steps

- **Maintain and update defensive publication record:** Continue to document all new embodiments, hardware revisions, and financial product innovations.
 - **Monitor prosecution and continuation activity:** Track any new filings or continuations by Samsung, IBM, or Hitachi that may attempt to broaden claim scope.
 - **Prepare for potential design-around by competitors:** Monitor for attempts to replicate BLEI-E's unique features and be prepared to assert defensive publications and trademark/copyright interests.
 - **Consider strategic claim drafting:** File additional claims or defensive publications covering any new system features or financial instruments not already protected.
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Conclusion

The BLEI-E system is unique in its integration of real-time, hardware-anchored biometric data capture, deterministic financial index construction, and regulated exchange-traded financial instruments. The prior art references-Samsung US20260037831A1, IBM US10346759B2, and Hitachi WO2023091144A1-do not anticipate, render obvious, or otherwise conflict with BLEI-E's core claim limitations or system features. The only conceptual overlap lies in the use of Bayesian/probabilistic inference, but the context, application, and technical implementation are fundamentally different. BLEI-E is well-positioned from both a freedom-to-operate and invalidity perspective, with robust defensive publication and IP protection strategies in place.

Claim Overlap Matrix Table (Attorney-Grade Summary)

BLEI-E Feature / Limitation	Samsung US20260037831 A1	IBM US10346759B 2	Hitachi WO2023091144 A1	BLEI-E Uniqueness / Notes
Real-time biometric data capture (Tech+Ball, wearables, stadium sensors)	No direct disclosure	Synthetic events from measured data (general)	Event detection from text data (not biometric)	BLEI-E uniquely claims hardware-anchored, real-time biometric capture and authentication
Athlete identity verification on contact (BioGrip Sensor)	Not disclosed	Not disclosed	Not disclosed	Unique to BLEI-E; not present in prior art
Immutable blockchain hashing of biometric data	Not disclosed	Not disclosed	Not disclosed	Unique to BLEI-E; no blockchain or audit trail in prior art
Multi-factor, normalized index construction (B-IVI, F-IVI, N-IVI, S-IVI)	No; meta-learning of distributions, not multi-factor index	Hierarchical event abstraction, not multi-factor index	Causal inference on events, not multi-factor index	BLEI-E's specific IVI framework and factor weighting are unique
Explicit factor weighting and signal smoothing (EWMA, Kalman)	No; Bayesian update, not explicit factor weighting	No; rule-based, not explicit weighting	No; Bayesian network, not explicit weighting	BLEI-E's quantitative weighting and smoothing protocols are unique
Stadium-wide biometric telemetry (Fan Velocity Score, S-IVI)	Not disclosed	Not disclosed	Not disclosed	Unique to BLEI-E; prior art does not address crowd biometric data
Exchange-traded financial	Not disclosed	Not disclosed	Not disclosed	BLEI-E uniquely claims regulated,

instruments (TET, PEI, SEI, NIL derivatives, ALT tokens)				exchange-traded financial products indexed to biometric data
Biometric-collateralized debt obligations (DBBI)	Not disclosed	Not disclosed	Not disclosed	Unique to BLEI-E; no debt instruments in prior art
Algorithmic trading, arbitrage, volatility overlays (BIM-APES)	Not disclosed	Not disclosed	Not disclosed	Unique to BLEI-E; prior art does not address trading overlays
Data integrity, spoofing prevention, SEC audit compliance	Not disclosed	Not disclosed	Not disclosed	Unique to BLEI-E; no comparable protocols in prior art
Defensive publication, copyright/trademark chain	Not disclosed	Not disclosed	Not disclosed	BLEI-E's IP protection strategy is unique
Bayesian/Probabilistic inference on data	Bayesian meta continual-learning and inference	Probabilistic inference from synthetic events	Causal inference using Bayesian networks	Overlap: All prior art use Bayesian/probabilistic inference, but not in BLEI-E's financial or biometric context
Event detection and abstraction	No; focuses on meta-learning, not event detection	Synthetic event abstraction from measured data	Event detection from text data, ad-hoc events	Partial overlap with BLEI-E's event-driven triggers, but BLEI-E's are hardware/biometric-based

Causal inference using Bayesian networks	No; Bayesian update, not causal networks	No; probabilistic inference, not causal	Yes; explicit use of Bayesian networks for causal inference	Overlap with BLEI-E's factor modeling, but BLEI-E applies it to financial index construction, not event forecasting
Meta-learning, neural network parameter training	Yes; meta-learning of neural networks	No; rule-based, not neural networks	No; event extraction uses deep learning, but not meta-learning	Samsung's meta-learning is not present in BLEI-E; BLEI-E uses deterministic, rule-based models

This report provides a structured, evidence-backed foundation for patent attorneys to assess BLEI-E's freedom to operate, invalidity risk, and strategic claim positioning relative to the cited prior art.

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